

1)Average sale price of houses

```
import numpy as np

# Dataset (Bedrooms, Sqft, Sale_Price)
house_data = np.array([
    [3, 1400, 750000],
    [5, 2200, 1250000],
    [4, 1800, 980000],
    [6, 3000, 1850000],
    [2, 1100, 520000]
])

# Houses with more than 4 bedrooms
filtered = house_data[house_data[:, 0] > 4]

# Average sale price
average_price = np.mean(filtered[:, 2])

print("Average Sale Price:", average_price)
```

Average Sale Price: 1550000.0

2)Logistic Regression

```
import numpy as np
from sklearn.linear_model import LogisticRegression

X = np.array([
    [1, 3, 120],
    [6, 15, 300],
    [0, 1, 80],
    [4, 10, 250],
    [2, 5, 160]
])

y = np.array([0, 1, 0, 1, 0])

model = LogisticRegression()
model.fit(X, y)

new_email = np.array([[2, 5, 160]])

prediction = model.predict(new_email)

if prediction[0] == 1:
    print("Spam")
else:
    print("Not Spam")
```

Not Spam

3)Logistic Regression model

```
import numpy as np
from sklearn.linear_model import LogisticRegression

X = np.array([
    [1, 3, 120],
    [6, 15, 300],
    [0, 1, 80],
    [4, 10, 250],
    [2, 5, 160]
])

y = np.array([0, 1, 0, 1, 0])

model = LogisticRegression()
model.fit(X, y)

new_email = np.array([[2, 5, 160]])

prediction = model.predict(new_email)

print(prediction)

if prediction[0] == 1:
    print("Spam")
else:
    print("Not Spam")
```

```
[0]
Not Spam
```

4)Linear Regression

```

import pandas as pd
from sklearn.linear_model import LinearRegression

data = {
    "Age_Years": [2, 5, 3, 7, 1],
    "Kms_Driven": [25000, 60000, 40000, 90000, 15000],
    "Engine_CC": [1200, 1500, 1300, 1800, 1000],
    "Resale_Price": [650000, 420000, 520000, 300000, 720000]
}

df = pd.DataFrame(data)

X = df[["Age_Years", "Kms_Driven", "Engine_CC"]]
y = df["Resale_Price"]

model = LinearRegression()
model.fit(X, y)

car_name = input("Enter Car Name: ")
age = int(input("Enter Car Age (Years): "))
kms = int(input("Enter Kilometers Driven: "))
engine = int(input("Enter Engine CC: "))

prediction = model.predict([[age, kms, engine]])

print("\nCar Name:", car_name)
print("Estimated Resale Price:", round(prediction[0], 2))

```

```

Enter Car Name: swift
Enter Car Age (Years): 4
Enter Kilometers Driven: 50000
Enter Engine CC: 1400

```

```

Car Name: swift
Estimated Resale Price: 495263.16

```

```

/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
  warnings.warn(

```

5) CART - Decision Tree Classifier

```

import pandas as pd
from sklearn.tree import DecisionTreeClassifier, export_text

data = {
    "Income": [45000, 30000, 60000, 28000, 52000],
    "Credit_Score": [720, 650, 780, 620, 710],
    "Existing_Loans": [0, 1, 0, 2, 1],
    "Approved": [1, 0, 1, 0, 1]
}

df = pd.DataFrame(data)

X = df[["Income", "Credit_Score", "Existing_Loans"]]
y = df["Approved"]

model = DecisionTreeClassifier(random_state=0)
model.fit(X, y)

income = int(input("Enter Income: "))
credit = int(input("Enter Credit Score: "))

```

```
credit = int(input("Enter Credit Score: "))
loans = int(input("Enter Existing Loans: "))

prediction = model.predict([[income, credit, loans]])

if prediction[0] == 1:
    print("Loan Status: Approved")
else:
    print("Loan Status: Rejected")

print("\nDecision Rules:")
print(export_text(model, feature_names=list(X.columns)))
```

```
Enter Income: 50000
Enter Credit Score: 700
Enter Existing Loans: 1
Loan Status: Approved
```

```
Decision Rules:
|--- Credit_Score <= 680.00
```